

# Computer Science & IT Engineering

## DPP 7; Top 250 Questions of DBMS Part- 7



**DBMS : DPP 7**

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# Normal Forms



## 15.3.5 Second Normal Form

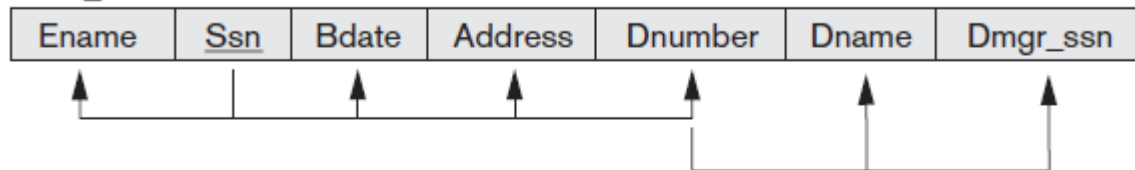
Second normal form (2NF) is based on the concept of *full functional dependency*. A functional dependency  $X \rightarrow Y$  is a **full functional dependency** if removal of any attribute  $A$  from  $X$  means that the dependency does not hold any more; that is, for any attribute  $A \in X$ ,  $(X - \{A\})$  does *not* functionally determine  $Y$ . A functional dependency  $X \rightarrow Y$  is a **partial dependency** if some attribute  $A \in X$  can be removed from  $X$  and the dependency still holds; that is, for some  $A \in X$ ,  $(X - \{A\}) \rightarrow Y$ . In Figure 15.3(b),  $\{\text{Ssn}, \text{Pnumber}\} \rightarrow \text{Hours}$  is a full dependency (neither  $\text{Ssn} \rightarrow \text{Hours}$  nor  $\text{Pnumber} \rightarrow \text{Hours}$  holds). However, the dependency  $\{\text{Ssn}, \text{Pnumber}\} \rightarrow \text{Ename}$  is partial because  $\text{Ssn} \rightarrow \text{Ename}$  holds.

**Definition.** A relation schema  $R$  is in 2NF if every nonprime attribute  $A$  in  $R$  is *fully functionally dependent* on the primary key of  $R$ .

Figure 15.3

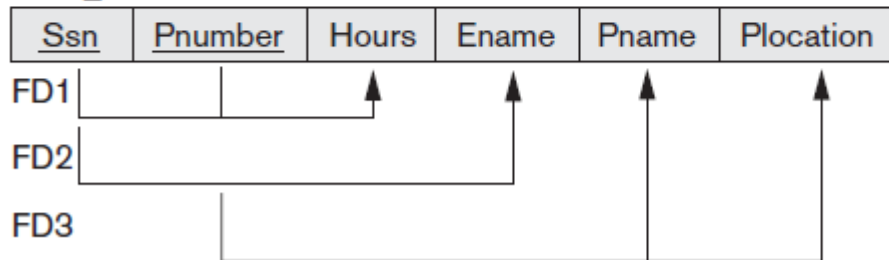
(a)

EMP\_DEPT



(b)

EMP\_PROJ



# Normal Forms

8.17 A functional dependency  $\alpha \rightarrow \beta$  is called a **partial dependency** if there is a proper subset  $\gamma$  of  $\alpha$  such that  $\gamma \rightarrow \beta$ . We say that  $\beta$  is *partially dependent* on  $\alpha$ . A relation schema  $R$  is in **second normal form** (2NF) if each attribute  $A$  in  $R$  meets one of the following criteria:

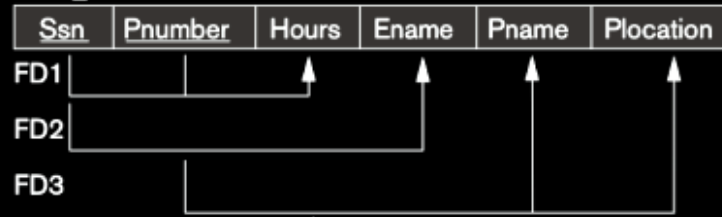
- It appears in a candidate key.
- It is not partially dependent on a candidate key.

# Normal Forms

## Second Normal Form

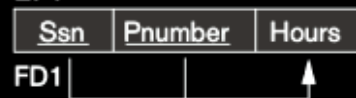
**Definition:** A relation schema R is in 2NF if every nonprime attribute A in R is fully functionally dependent on the primary key of R.

EMP\_PROJ

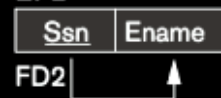


2NF Normalization

EP1

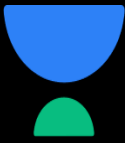


EP2



EP3





Let  $R(A, B, C, D, E, P, G)$  be a relational schema in which the following functional dependencies are known to hold:

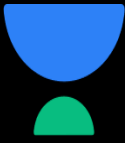
$AB \rightarrow CD$ ,  $DE \rightarrow P$ ,  $C \rightarrow E$ ,  $P \rightarrow C$  and  $B \rightarrow G$ .

The relational schema  $R$  is

- A** In BCNF
- B** In 3NF, but not in BCNF
- C** In 2NF, but not in 3NF
- D** Not in 2NF



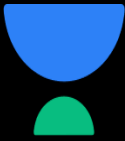
# Normal Forms



## Third Normal Form

**Definition:** According to Codd's original definition, a relation schema  $R$  is in 3NF if it satisfies 2NF and no nonprime attribute of  $R$  is transitively dependent on the primary key.

**Definition:** A relation schema  $R$  is in third normal form (3NF) if, whenever a nontrivial functional dependency  $X \rightarrow A$  holds in  $R$  either (a)  $X$  is a superkey of  $R$ , or (b)  $A$  is a prime attribute of  $R$ .



### EMP\_DEPT

| Ename | <u>Ssn</u> | Bdate | Address | Dnumber | Dname | Dmgr_ssn |
|-------|------------|-------|---------|---------|-------|----------|
| ↑     |            | ↑     | ↑       | ↑       | ↑     | ↑        |

### 3NF Normalization

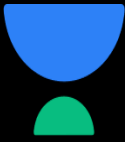
#### ED1

| Ename | <u>Ssn</u> | Bdate | Address | Dnumber |
|-------|------------|-------|---------|---------|
| ↑     |            | ↑     | ↑       | ↑       |

#### ED2

| <u>Dnumber</u> | Dname | Dmgr_ssn |
|----------------|-------|----------|
| ↑              |       | ↑        |

# Boyce – Codd Normal Form



**Definition:** A relation schema  $R$  is in BCNF if whenever a nontrivial functional dependency  $X \rightarrow A$  holds in  $R$ , then  $X$  is a superkey of  $R$ .



# Normal Forms

## Second Normal Form

**Definition:** A relation schema R is in 2NF if every nonprime attribute A in R is fully functionally dependent on the primary key of R.

EMP\_PROJ

| <u>Ssn</u> | <u>Pnumber</u> | Hours | Ename | Pname | Plocation |
|------------|----------------|-------|-------|-------|-----------|
|------------|----------------|-------|-------|-------|-----------|



2NF Normalization

EP1

| <u>Ssn</u> | <u>Pnumber</u> | Hours |
|------------|----------------|-------|
|------------|----------------|-------|



EP2

| <u>Ssn</u> | Ename |
|------------|-------|
|------------|-------|



EP3

| <u>Pnumber</u> | Pname | Plocation |
|----------------|-------|-----------|
|----------------|-------|-----------|



Q.4

R(ABCDE) { $AB \rightarrow D$ ,  $C \rightarrow B$ ,  $E \rightarrow A$ } Identify the Partial FD's & Checking 2NF?



Q.5

$R(ABCDE)$   $\{AB \rightarrow CDE, B \rightarrow C\}$  Identify the Partial FD's & Checking 2NF?



Q.6

$R(ABCD) \{AB \rightarrow CD, CD \rightarrow AB, B \rightarrow D\}$  Identify the 2NF?



Q.7

$R(ABCD) \{AB \rightarrow CD, CD \rightarrow AB, A \rightarrow C\}$  Identify the 2NF?



# Normal Forms



## Third Normal Form

**Definition:** According to Codd's original definition, a relation schema  $R$  is in 3NF if it satisfies 2NF and no nonprime attribute of  $R$  is transitively dependent on the primary key.

**Definition:** A relation schema  $R$  is in third normal form (3NF) if, whenever a nontrivial functional dependency  $X \rightarrow A$  holds in  $R$  either (a)  $X$  is a superkey of  $R$ , or (b)  $A$  is a prime attribute of  $R$ .

Q.8

$R(ABCD) \{AB \rightarrow CD, CD \rightarrow AB, A \rightarrow C\}$  Identify  $AB \rightarrow C$  is the Transitive FD & R is in 3NF or Not ?



Q.9

$R(ABCD) \{AB \rightarrow CD, CD \rightarrow AB, A \rightarrow C\}$  Show Partial FD may be Transitive or Not ?





**Q.10**

$R(ABC) \{A \rightarrow B, B \rightarrow C\}$  Check  $A \rightarrow C$  Partial FD & Transitive or Not?



## [MCQ]

Consider a relation R (P, Q, R, S, T) with the functional dependency's as :

{

$PQR \rightarrow ST$

$S \rightarrow PQ$

}

Which of the following is the total number of superkeys for the above relation R?

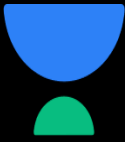
(a) 10

(b) 12

(c) 8

(d) 6

[NAT]

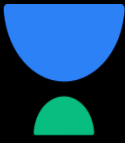


Consider the following relational schema  $R(ABCDE)$  with functional dependencies:

$\{AB \rightarrow C, D \rightarrow E, E \rightarrow A, D \rightarrow B\}$

The number of additional relations are required to convert it into lossless join & Dependency Presenting 2NF Decomposition is-

## [NAT]



Consider a relation R with four Attributes A,B,C & D with the following FD's.

FD-I  $C \rightarrow D, C \rightarrow A, B \rightarrow C$

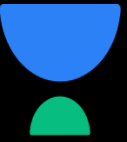
FD-II  $ABC \rightarrow D, D \rightarrow A$

FD-III  $A \rightarrow B, BC \rightarrow D, A \rightarrow C$

FD-IV  $AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B$

How many above FD are satisfied the 3NF (Third Normal Form) for a relation schema R(ABCD) is\_\_?

## [NAT]



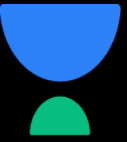
FD-I  $C \rightarrow D, C \rightarrow A, B \rightarrow C$

FD-II  $ABC \rightarrow D, D \rightarrow A$

FD-III  $A \rightarrow B, BC \rightarrow D, A \rightarrow C$

FD-IV  $AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B$

## [NAT]



FD-I  $C \rightarrow D, C \rightarrow A, B \rightarrow C$

FD-II  $ABC \rightarrow D, D \rightarrow A$

FD-III  $A \rightarrow B, BC \rightarrow D, A \rightarrow C$

FD-IV  $AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B$

## [MCQ]



Assume a relation  $R = (P, Q, R, S)$  and a set  $F$  of functional dependencies:

$F = \{PR \rightarrow S, S \rightarrow P, S \rightarrow Q, S \rightarrow R\}$ , Highest normal form satisfied by the relation  $R$  is?

- A. 2NF
- B. 3NF
- C. BCNF
- D. 1NF

[MSQ]



Consider the following relation  $R(P, Q, R, S)$  and functional dependencies  $F$  that hold over the relation

$F = \{P \rightarrow QS, RS \rightarrow Q, R \rightarrow S, Q \rightarrow PS\}$ . The relation  $R$  is in which of the following normal form?

- A. 1NF
- B. 2NF
- C. 3NF
- D. BCNF

[MCQ]

Suppose functional dependency  $Q \rightarrow R$  holds in relation R (P, Q, R, S) which additional FD will make R be in 3NF, but not BCNF?

- A.  $S \rightarrow PQ$
- B.  $PR \rightarrow S$
- C.  $RS \rightarrow Q$
- D.  $PS \rightarrow Q$



## [MCQ]



Consider the following relation with given functional dependencies.

$R(ABCDEFGH)$  with FD set  $F$

$F: [AB \rightarrow CD, D \rightarrow EG, F \rightarrow H, C \rightarrow EF, H \rightarrow A, G \rightarrow B, A \rightarrow B]$

The relation is in which of the highest normal form?

A 1 NF

B 2 NF

C 3 NF

D BCNF

[MCQ]

Assume a relation schema R with 5 attributes P, Q, R, S, T and the set of FD'S  $P \rightarrow RS$ ,  $Q \rightarrow RT$ ,  $T \rightarrow Q$  consider the statements:

**S<sub>1</sub>:** The only candidate key of R is PQ and PT

**S<sub>2</sub>:** The highest normal form satisfied by R is 2NF

Which of the statement is true?

- A. Only S<sub>1</sub> is true
- B. Only S<sub>2</sub> is true
- C. Both S<sub>1</sub> and S<sub>2</sub> are true
- D. Neither S<sub>1</sub> nor S<sub>2</sub> is true



# *Video Solution of DPP 7*



**Link:** <https://unacademy.com/a/Top-250-Questions-on-DBMS.html%27>

# *Video Solution of DPP 7 LINK*



# THANK YOU!

Here's to a cracking journey ahead!